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Rajshahi University of Engineering & Technology

Department of Electrical & Electronic Engineering

Project Report

"Arduino-Based Ultrasonic Radar System for Real-Time Object Detection"

Course No. : EEE 3100

Course Title : Electronics Shop Practice

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1. Project in Brief

Date of Project Assignment		20 May, 2026
Date of Submission		14 June, 2026
Title of the Project	:	Arduino-Based Ultrasonic Radar System for Real-Time Object Detection
Project Financed by	:	Self
Total Cost of Project Completion	:	1130 BDT

2. Project Background:

The rapid advancement of embedded systems and sensor fusion technologies has created new opportunities for developing low-cost surveillance and object detection systems that were previously achievable only with expensive radio-frequency hardware. Conventional RADAR (Radio Detection And Ranging) systems operate on microwave frequencies in the GHz range and require specialized RF components, antenna arrays, and signal processing hardware that remain inaccessible at the undergraduate prototyping level. Ultrasonic-based radar systems offer a practical and educationally significant alternative, enabling real-time spatial detection, angular sweep coverage, and visual display using widely available, low-cost microcontroller platforms.

Existing ultrasonic radar implementations in literature and open-source repositories primarily demonstrate static distance measurement and basic 2D polar sweep visualization. These implementations do not address the velocity estimation of detected moving objects or the classification of targets based on their kinematic signatures. This project extends the conventional single-sensor ultrasonic radar architecture by incorporating a dual-sensor velocity estimation subsystem using a time-differencing algorithm, a microwave Doppler pre-trigger module (RCWL-0516) for wide-range early warning, and a discrete Kalman filter for angular noise suppression.

The proposed system integrates an SG90 servo-mounted HC-SR04 ultrasonic sensor for angular sweep radar coverage, a second fixed HC-SR04 sensor for time-of-flight based velocity computation, and an RCWL-0516 3.18 GHz Doppler module as a long-range motion detector. All sensor data is processed on an Arduino Uno microcontroller and transmitted to a Processing IDE visualization interface on a PC via serial communication. The GUI renders a professional polar radar display with persistence trails, real-time velocity readout, color-coded object classification labels, and proximity alert functionality — closely mirroring the visual and functional language of real-world radar systems.

3. Objectives :

1. To design and implement a servo-mounted ultrasonic radar system capable of a 150-degree angular sweep with real-time polar coordinate object detection and visualization on a PC display.
2. To implement a dual HC-SR04 time-differencing velocity estimation algorithm that computes the approach speed of detected objects in real time using the formula $v = \Delta d / \Delta t$.
3. To integrate an RCWL-0516 microwave Doppler module as a wide-range motion pre-trigger, providing early-warning detection independent of the directional ultrasonic sensors.

4. To apply a discrete Kalman filter on the servo angle measurement to suppress mechanical jitter and improve the angular positioning accuracy of the radar plot.
5. To develop a velocity-threshold-based object classification system that categorizes detected objects into kinematic classes: STATIC, SLOW, BIRD-equivalent, and VEHICLE-equivalent.
6. To build an advanced Processing IDE radar GUI featuring a persistence trail effect, color-coded classification labels, live HUD readout of angle, distance, velocity and RCWL state, and a proximity alert flash triggered at 15 cm.

4. Methodology to be Adopted in the Project:

The project is implemented in three integrated parts: mechanical assembly, hardware electronics, and software development.

Hardware:

1. The primary sweep sensor (HC-SR04 #1) is mounted on the SG90 servo horn using a mechanical bracket and sweeps continuously from 15 to 165 degrees in 1-degree increments. The servo receives PWM control signals from Arduino digital pin D12, with a 25 ms dwell at each step to allow mechanical settling before the ultrasonic pulse fires.
2. A second HC-SR04 (fixed at 90 degrees on the breadboard) functions as the velocity baseline sensor. It measures the distance to any approaching object at each loop iteration. Consecutive distance readings and their elapsed timestamps are used to compute approach velocity: $v = (d_{prev} - d_{curr}) / \Delta t$, expressed in m/s.
3. The RCWL-0516 microwave Doppler sensor is connected to Arduino interrupt pin D2. When any moving object enters its detection field (up to 7 m, 360-degree coverage), its OUT pin goes HIGH, triggering a motion alert flag visible in the Processing GUI.
4. All sensors share the 5V and GND rails on the breadboard. HC-SR04 #1 uses TRIG/ECHO on D10/D11; HC-SR04 #2 uses D6/D7; SG90 signal wire connects to D12; RCWL-0516 OUT connects to D2.

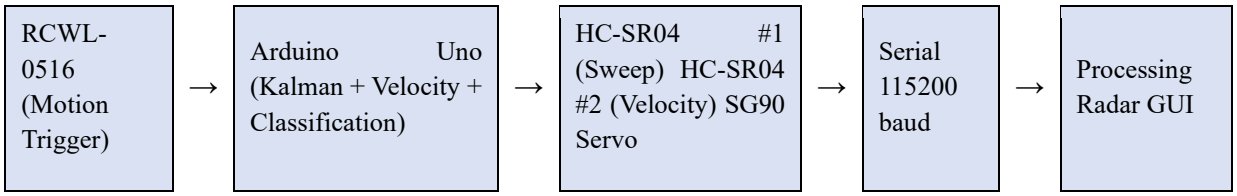
Arduino Software:

1. The main control loop executes a continuous forward (15°→165°) and backward (165°→15°) servo sweep. At each 1-degree step, the sweep sensor fires, and the fixed velocity sensor fires independently to update the velocity estimate.
2. A discrete Kalman filter with process noise Q = 0.01 and measurement noise R = 0.1 runs at every servo step, producing a smoothed angular estimate from the raw mechanical servo position to reduce jitter-induced dot scatter on the radar plot.
3. The classify() function applies velocity thresholds to categorize detected objects: below 0.3 m/s = STATIC, 0.3–0.8 m/s = SLOW, 0.8–2.5 m/s = BIRD, above 2.5 m/s = VEHICLE. These thresholds are tunable for environment calibration.
4. Each data frame is serialized at 115200 baud in the format: smoothed_angle, distance, velocity, rcwl_state, classification_label — five comma-separated fields terminated with a newline, parsed by the Processing sketch in real time.

Processing IDE GUI:

1. The GUI renders a semicircular polar grid with four concentric range rings at 10 cm intervals (10–40 cm) and radial reference lines every 30 degrees with angular labels, on a dark background.
2. A green sweep line follows the live Kalman-filtered servo angle. Detected objects within range are rendered as filled circles at their polar coordinates, color-coded by classification: yellow = BIRD, red = VEHICLE, cyan = SLOW, grey = STATIC, green = unclassified.
3. An 80-frame trail buffer stores recent detection coordinates with alpha-fading to produce a radar persistence afterglow effect, closely matching the appearance of professional plan-position indicator (PPI) radar displays.
4. A HUD panel (top-left) displays live angle, distance, velocity, RCWL state, and classification label. A motion badge appears top-right when RCWL fires. A full-screen red flash alert triggers when any object enters the 15 cm proximity zone.

Block Diagram:



5. Expected Results:

Upon successful completion, the system is expected to:

1. Sweep the full 150-degree arc continuously and render a real-time polar radar display in the Processing GUI with smooth angular motion and an angular error of less than 2 degrees after Kalman filtering.
2. Detect objects within the 2–40 cm range on the sweep sensor with a distance accuracy of approximately ±3 mm, consistent with HC-SR04 datasheet specifications at 5V supply.
3. Estimate the approach velocity of objects moving toward the fixed baseline sensor with sufficient resolution to reliably distinguish STATIC, SLOW, BIRD-equivalent, and VEHICLE-equivalent kinematic classes within the 0–5 m/s range.
4. Trigger the RCWL-0516 motion alert consistently for any moving object within a 5–7 meter radius, demonstrating wide-area pre-trigger capability that is independent of the directional ultrasonic sweep.
5. Display the correct color-coded classification label and velocity readout in the GUI HUD for each detected object with no misclassification during stationary tabletop testing.
6. Activate the red proximity alert flash reliably whenever any detected object enters the 15 cm danger zone, demonstrating the system's applicability as a short-range intrusion detection prototype.



Fig: Object Detection by Arduino-Based Ultrasonic Radar System

6. Financial Statement:

Sl. No.	**Name of the Item**	**Specification**	**Qty.**	**Unit Price (BDT)**	**Total Cost (BDT)**
1	Arduino Uno R3	ATmega328P, 16 MHz, USB	1	500	500
2	HC-SR04 Ultrasonic Sensor	40 kHz, 2–400 cm, 5V	2	60	120
3	SG90 Servo Motor	180°, PWM, 4.8–6V	1	120	120
4	RCWL-0516 Doppler Module	3.18 GHz, 5–7 m range	1	80	80
5	Breadboard	830 tie-points	1	120	120
6	Jumper Wires (M-M)	20 cm, assorted	1 set	80	80
7	USB Cable Type-B	Arduino programming	1	60	60
8	Mounting Bracket / Tape	Sensor-servo mechanical mount	1 set	50	50
Grand Total					**1,130 BDT**

7. Project Timeframe:

The following Gantt chart presents the tentative week-wise timeline for the project:

Task	W1	W2	W3	W4	W5	W6	W7	W8
Literature Review	■	■						
Component Procurement	■	■						
Hardware Assembly & Wiring		■	■					
Arduino Code Development			■	■				
Kalman Filter & Velocity Algorithm				■	■			
Processing GUI Development				■	■			
System Integration & Testing					■	■		
Performance Evaluation						■	■	
Report Writing							■	■

8. Addressing Complex Engineering Problems (P's):

This project addresses the following Complex Engineering Problem attributes:

Attributes	Justification
P1 (Depth of Knowledge)	The project demands integrated knowledge spanning embedded systems programming, digital signal processing (discrete Kalman filtering), sensor physics (ultrasonic pulse-echo and microwave Doppler principles), serial communication, real-time graphical rendering, and kinematic classification logic — crossing multiple sub-disciplines of electrical and electronic engineering simultaneously.
P2 (Range of Conflicting Requirements)	A fundamental tradeoff exists between radar sweep speed and measurement accuracy: faster servo stepping reduces dwell time and degrades distance reliability, while slower sweep reduces temporal resolution for moving object tracking. Simultaneously, the fixed-sensor velocity estimation accuracy competes with sweep sensor timing, requiring careful loop synchronization to avoid cross-trigger interference between the two HC-SR04 modules.

P4 (Familiarity of Issues)	Practical challenges encountered include ultrasonic cross-talk when both sensors fire in close temporal proximity, servo mechanical backlash introducing angular scatter on the radar plot, RCWL-0516 false triggering from fluorescent lighting interference, and Processing IDE serial buffer latency at high baud rates — all representative of real-world sensor fusion and embedded systems integration problems.
P5 (Applicable Codes and Standards)	The project adheres to Arduino C++ programming conventions, standard UART serial protocol (115200 baud, 8N1), PWM servo control standards (20 ms period, 1–2 ms pulse width), and HC-SR04 and RCWL-0516 datasheet operating specifications. The Kalman filter follows the standard discrete-time recursive estimation formulation as described by Kalman (1960).

9. References:

[1] R. E. Kalman, "A New Approach to Linear Filtering and Prediction Problems," Trans. ASME – J. Basic Eng., vol. 82, ser. D, pp. 35–45, 1960.

[2] J. Borenstein and Y. Koren, "Real-Time Obstacle Avoidance for Fast Mobile Robots," IEEE Trans. Syst., Man, Cybern., vol. 19, no. 5, pp. 1179–1187, Sep./Oct. 1989.